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About CertyIQ

We here at CertyIQ eventually got enough of the industry's greedy exam paid for. Our team of IT professionals comes with years of experience in the IT industry Prior to training CertyIQ we worked in test areas where we observed the horrors of the paywall exam preparation system.

The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

Doubt Support

We have developed a very scalable solution using which we are able to solve 400+ doubts every single day with an average rating of 4.8 out of 5.

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John

October 19, 2022



Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022



Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today - 3 September with 942 score.

Microsoft

(DP-900)

Microsoft Azure Data Fundamentals

Total: **256 Questions**

Link: <https://certyiq.com/papers?provider=microsoft&exam=dp-900>

Question: 1

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Descriptive analytics tells you

	▼
what is most likely to occur in the future.	
what occurred in the past.	
which actions you can perform to affect outcomes.	
why something occurred in the past.	

Answer:**Answer Area**

Descriptive analytics tells you

	▼
what is most likely to occur in the future.	
what occurred in the past.	
which actions you can perform to affect outcomes.	
why something occurred in the past.	

Explanation:

Descriptive analytics tells "What happened in the past". So B is the right choice.

Reference:

<https://demand-planning.com/2020/01/20/the-differences-between-descriptive-diagnostic-predictive-cognitive-analytics/>**Question: 2**

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Normalization involves eliminating relationships between database tables.	☐	☐
Normalizing a database reduces data redundancy.	☐	☐
Normalization improves data integrity.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Normalization involves eliminating relationships between database tables.	☐	☒
Normalizing a database reduces data redundancy.	☒	☐
Normalization improves data integrity.	☒	☐

Explanation:

Normalization is the process of organizing data in a database. This includes creating tables and establishing relationships between those tables according to rules designed both to protect the data and to make the database more flexible by eliminating redundancy and inconsistent dependency.

Reference:

<https://docs.microsoft.com/en-us/office/troubleshoot/access/database-normalization-description#:~:text=Normalization%20is%20the%20process%20of,eliminating%20redundancy%20and%20inconsi>

<https://www.sqlshack.com/what-is-database-normalization-in-sql-server/>

Question: 3

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

An extract, transform, and load (ETL) process requires

- a matching schema in the data source and the data target.
- a target data store powerful enough to transform data.
- data that is fully processed before being loaded to the target data store.
- that the data target be a relational database.

Answer:

Answer Area

An extract, transform, and load (ETL) process requires

- a matching schema in the data source and the data target.
- a target data store powerful enough to transform data.
- data that is fully processed before being loaded to the target data store.
- that the data target be a relational database.

Explanation:

Extract, transform, and load (ETL) is a data pipeline used to collect data from various sources, transform the data according to business rules, and load it into a destination data store.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/etl>

Question: 4

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

In batch processing,

- data is always inserted one row at a time.
- data is processed in real-time.
- latency in delivering data processing results is acceptable.
- processing can only execute serially.

Answer:

Answer Area

In batch processing,

- data is always inserted one row at a time.
- data is processed in real-time.
- latency in delivering data processing results is acceptable.
- processing can only execute serially.

Explanation:

When to use batch processing.

You might expect latencies when using batch processing. For many situations, however, this type of delay before the transfer of data begins is not a big issue - the processes that use this function are not mission critical at the exact moment.

Reference:

<https://www.bmc.com/blogs/what-is-batch-processing-batch-processing-explained/>

Question: 5

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Transcribing audio files is an example of

- cognitive
- descriptive
- predictive
- prescriptive

analytics.

Answer:

Answer Area

Transcribing audio files is an example of

- cognitive
- descriptive
- predictive
- prescriptive

analytics.

Explanation:

Transcribe means converting from Audio to Text. This is similar to Alexa, Alexa is a Cognitive device. Cognitive is the correct answer.

Reference:

<https://azure.microsoft.com/en-us/services/cognitive-services/speech-services/>

Question: 6

CertyIQ

DRAG DROP -

Match the types of analytics that can be used to answer the business questions.

To answer, drag the appropriate analytics type from the column on the left to its question on the right. Each analytics type may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Analytics Types

Answer Area

- Cognitive
- Diagnostic
- Descriptive
- Predictive
- Prescriptive

- Why did sales increase last month?
- How do I allocate my budget to buy different inventory items?
- Which people are mentioned in a company's business documents?

Answer:

Analytics Types

Answer Area

- Cognitive
- Diagnostic
- Descriptive
- Predictive
- Prescriptive

- Diagnostic Why did sales increase last month?
- Prescriptive How do I allocate my budget to buy different inventory items?
- Descriptive Which people are mentioned in a company's business documents?

Explanation:

Box 1: Diagnostic -

Diagnostic Analytics: At this stage you can begin to answer some of those why questions. Historical data can begin to be measured against other data to answer the question of why something happened in the past. This is the process of gathering and interpreting different data sets to identify anomalies, detect patterns, and determine relationships.

Box 2: Prescriptive -

Prescriptive analytics is a combination of data, mathematical models, and various business rules to infer actions to influence future desired outcomes.

Predictive analytics, broadly speaking, is a category of business intelligence that uses descriptive and

predictive variables from the past to analyze and identify the likelihood of an unknown future outcome

Box 3: Descriptive -

- ⇒ Generally speaking, data analytics comes in four types:
- ⇒ Descriptive, to answer the question: What's happening?
- ⇒ Diagnostic, to answer the question: Why's happening?
- ⇒ Predictive, to answer the question: What will happen?
- ⇒ Prescriptive, to answer the question: What actions should we take?



Reference:

<https://demand-planning.com/2020/01/20/the-differences-between-descriptive-diagnostic-predictive-cognitive-analytics/> <https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/>

Question: 7

CertyIQ

HOTSPOT -

You have the following JSON document.


```
"customer" : {  
  "first name" : "Ben",  
  "last name" : "Smith",  
  "address" : {  
    "line 1" : "161 Azure Ln",  
    "line 2" : "Palo Alto",  
    "ZIP code" : "54762"  
  },  
  "social media": [  
    {  
      "service" : "twitter",  
      "handle" : "@bensmith"  
    },  
    {  
      "service" : "linkedin",  
      "handle" : "bensmith"  
    }  
  ],  
  "phone numbers": [  
    {  
      "type" : "mobile",  
      "number" : "555-555-555"  
    }  
  ]  
}
```

Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the JSON document.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Customer is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Address is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Social media is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Answer:

Answer Area

Customer is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Address is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Social media is [answer choice].

	▼
a nested array	
a nested object	
a root object	

Explanation:

Reference:

https://www.w3schools.com/js/js_json_arrays.asp

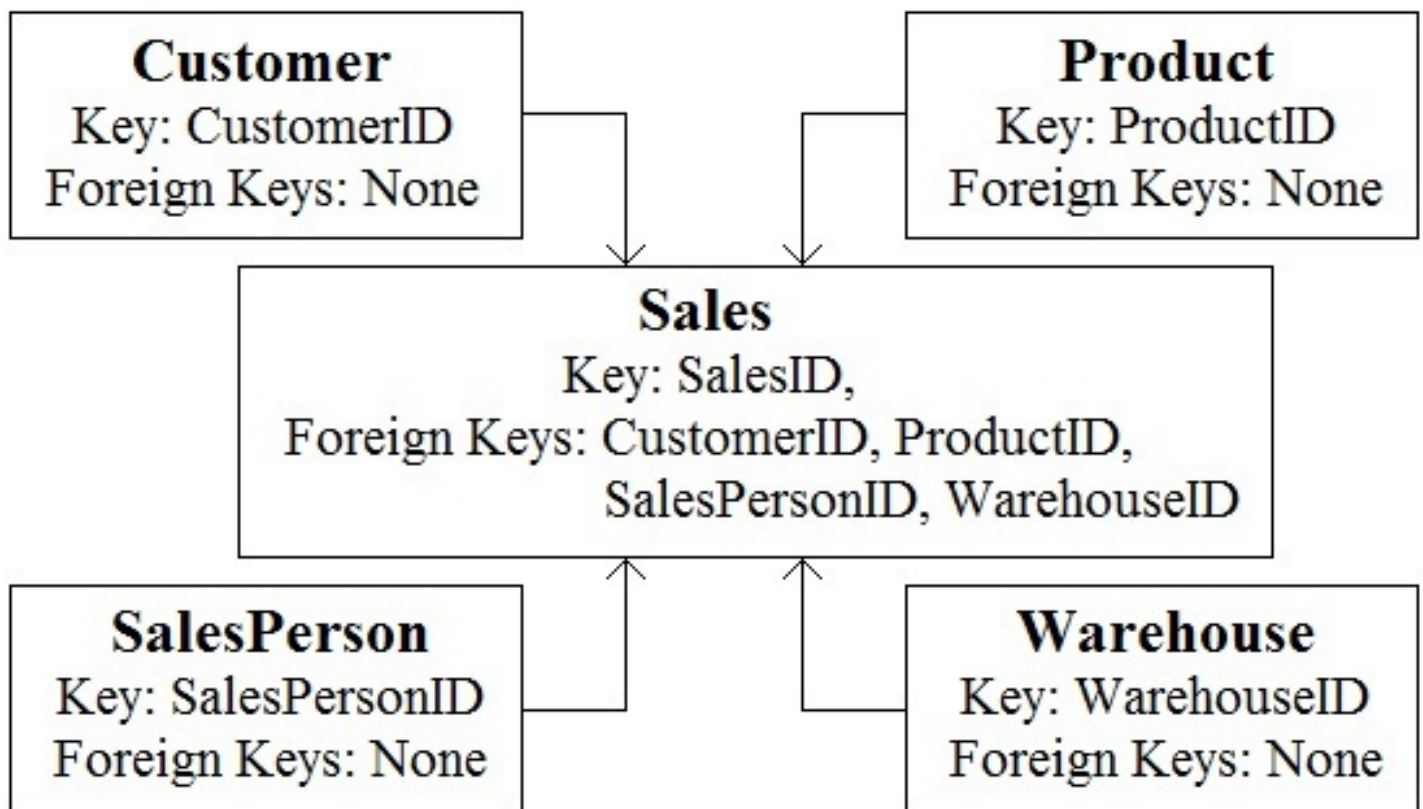
https://www.w3schools.com/js/js_json_objects.asp

Question: 8

CertyIQ

HOTSPOT -

You are reviewing the data model shown in the following exhibit.



Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.

NOTE: Each correct selection is worth one point

Hot Area:

Answer Area

The data model is a [answer choice].

	▼
transactional model	
star schema	
snowflake schema	

Customer is a [answer choice] table.

	▼
fact	
dimension	
bridge	

Answer:

Answer Area

The data model is a [answer choice].

transactional model
star schema
snowflake schema

Customer is a [answer choice] table.

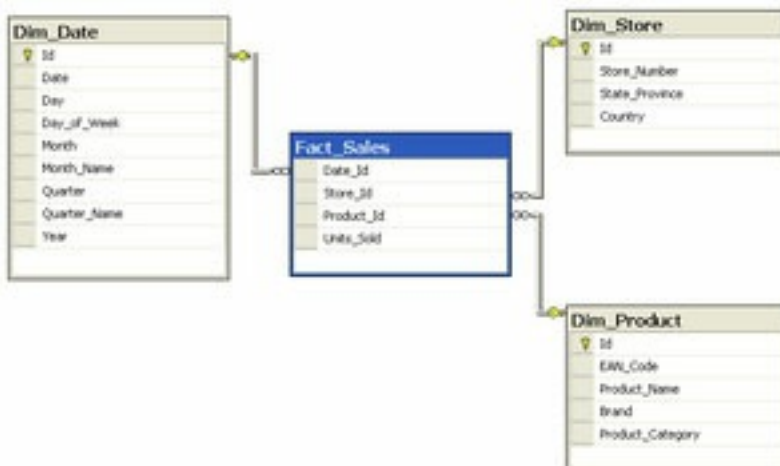
fact
dimension
bridge

Explanation:

Box 1: star schema -

In computing, the star schema is the simplest style of data mart schema and is the approach most widely used to develop data warehouses and dimensional data marts. The star schema consists of one or more fact tables referencing any number of dimension tables. The star schema is an important special case of the snowflake schema, and is more effective for handling simpler queries.

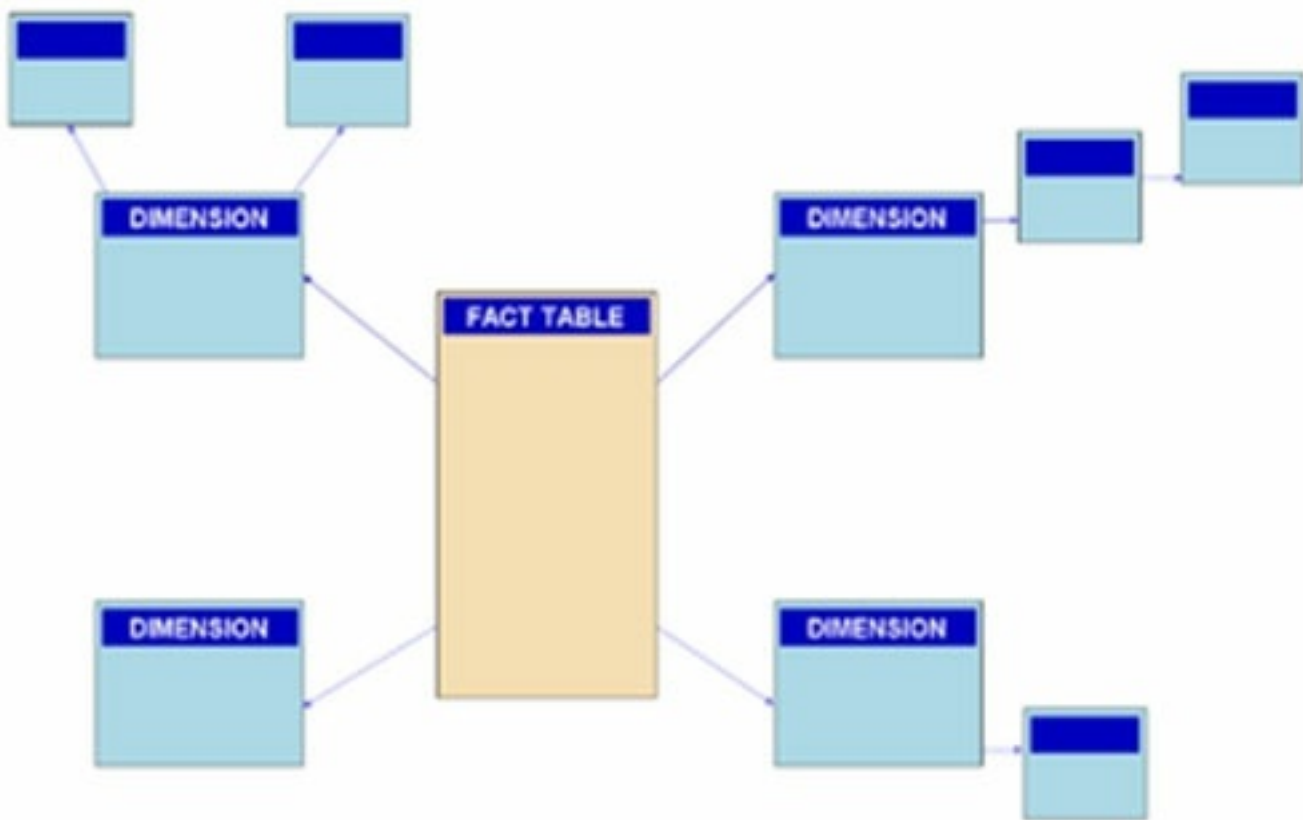
Example:



Incorrect Answers:

The data in the question is not normalized.

The snowflake schema is a variation of the star schema, featuring normalization of dimension tables. Example:



Note: A snowflake schema is a logical arrangement of tables in a multidimensional database such that the entity relationship diagram resembles a snowflake shape. The snowflake schema is represented by centralized fact tables which are connected to multiple dimensions.[citation needed]. "Snowflaking" is a method of normalizing the dimension tables in a star schema. When it is completely normalized along all the dimension tables, the resultant structure resembles a snowflake with the fact table in the middle.

Box 2: dimension -

The star schema consists of one or more fact tables referencing any number of dimension tables.

Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-tables-overview> https://en.wikipedia.org/wiki/Star_schema https://en.wikipedia.org/wiki/Snowflake_schema <https://azure.microsoft.com/en-us/blog/data-models-within-azure-analysis-services-and-power-bi/>

Question: 9

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

The massively parallel processing (MPP) engine of Azure

Synapse Analytics

- | | |
|--|---|
| | ▼ |
| distributes processing across compute nodes. | |
| distributes processing across control nodes. | |
| redirects client connections across compute nodes. | |
| redirects client connections across control nodes. | |

Answer:

Answer Area

The massively parallel processing (MPP) engine of Azure Synapse Analytics

- distributes processing across compute nodes.
- distributes processing across control nodes.
- redirects client connections across compute nodes.
- redirects client connections across control nodes.

Explanation:

Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/massively-parallel-processing-mpp-architecture>

Question: 10

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HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

- A clustered index
- A FileTable
- A foreign key
- A stored procedure

is an object associated with a table that sorts and stores the data rows in the table based on their key values.

Answer:

Answer Area

- A clustered index
- A FileTable
- A foreign key
- A stored procedure

is an object associated with a table that sorts and stores the data rows in the table based on their key values.

Explanation:

Clustered indexes sort and store the data rows in the table or view based on their key values. These are the columns included in the index definition. There can be only one clustered index per table, because the data rows themselves can be stored in only one order.

Reference:

<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/clustered-and-nonclustered-indexes-described?view=sql-server-ver15>

Question: 11

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A relational database is appropriate for scenarios that involve a high volume of

	▼
changes to relationships between entities	
geographically distributed writes	
transactional writes	
writes that have varying data structures	

Answer:

Answer Area

A relational database is appropriate for scenarios that involve a high volume of

	▼
changes to relationships between entities	
geographically distributed writes	
transactional writes	
writes that have varying data structures	

Explanation:

Disadvantages of non-relational databases include: Data Consistency " non-relational databases do not perform ACID transactions.

Note: Relational databases are optimized for writes. They are optimized for consistency and availability.

Advantages of relational databases include simplicity, ease of data retrieval, data integrity, and flexibility.

Incorrect Answers:

Use a relational database when data that you work with is structured, and the structure is not subject to frequent changes.

Use Cloud storage (no relational database) for geographically distributed writes.

Reference:

<https://towardsdatascience.com/choosing-the-right-database-c45cd3a28f77>

Question: 12

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.
Hot Area:

Answer Area

Statements	Yes	No
Batch processing can output data to a file store	☐	☐
Batch processing can output data to a relational database	☐	☐
Batch processing can output data to a NoSQL database	☐	☐

Answer:

Answer Area

Statements	Yes	No
Batch processing can output data to a file store	☒	☐
Batch processing can output data to a relational database	☒	☐
Batch processing can output data to a NoSQL database	☒	☐

Explanation:

Box 1: Yes -

Big data solutions often use long-running batch jobs to filter, aggregate, and otherwise prepare the data for analysis. Usually these jobs involve reading source files from scalable storage (like HDFS, Azure Data Lake Store, and Azure Storage), processing them, and writing the output to new files in scalable storage.

Box 2: Yes-

Box 3: Yes-

Batch processing can have output to SQL Database, Hive, HBase, Spark SQL. Look at the link and the diagram there

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/big-data/batch-processing>

Question: 13

DRAG DROP -
Your company plans to load data from a customer relationship management (CRM) system to a data warehouse by

using an extract, load, and transform (ELT) process.

Where does data processing occur for each stage of the ELT process? To answer, drag the appropriate locations to the correct stages. Each location may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Select and Place:

Locations

An in-memory data integration tool

The CRM system

The data warehouse

Answer Area

Extract:

Location

Load:

Location

Transform:

Location

Answer:

Locations

An in-memory data integration tool

The CRM system

The data warehouse

Answer Area

Extract:

The CRM system

Load:

The data warehouse

Transform:

The data warehouse

Explanation:

For ELT, the correct answer is CRM-DWH-DWH

Box 1: The CRM system -

Data is extracted from the CRM system.

Box 2: The data warehouse -

Data is loaded to the data warehouse.

Box 3: The data warehouse -

<https://learn.microsoft.com/en-us/azure/architecture/data-guide/relational-data/elt>

Extract, load, and transform (ELT) differs from ETL solely in where the transformation takes place. In the ELT pipeline, the transformation occurs in the target data store. Instead of using a separate transformation engine, the processing capabilities of the target data store are used to transform data.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/elt>

Question: 14

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A visualization that shows a university's current student enrollment versus the maximum capacity is an example of

	▼
cognitive	
descriptive	
predictive	
prescriptive	

 analytics.

Answer:

Answer Area

A visualization that shows a university's current student enrollment versus the maximum capacity is an example of

	▼
cognitive	
descriptive	
predictive	
prescriptive	

 analytics.

Explanation:

Generally speaking, data analytics comes in four types (Figure 1):

Descriptive, to answer the question: What's happening?

Diagnostic, to answer the question: Why's happening?

Predictive, to answer the question: What will happen?

Prescriptive, to answer the question: What actions should we take?

Reference:

<https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/>

Question: 15

DRAG DROP -

Match the types of visualizations to the appropriate descriptions.

To answer, drag the appropriate visualization type from the column on the left to its description on the right. Each visualization type may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Visualizations

Key influencer

Scatter

Treemap

Answer Area

Visualization

A chart of colored, nested rectangles that displays individual data points represented by the size and color of a relative rectangle.

Visualization

A chart that displays the major contributors of a selected result or value.

Visualization

A chart that shows the relationship between two numerical values.

Answer:

Visualizations

Key influencer

Scatter

Treemap

Answer Area

Treemap

A chart of colored, nested rectangles that displays individual data points represented by the size and color of a relative rectangle.

Key influencer

A chart that displays the major contributors of a selected result or value.

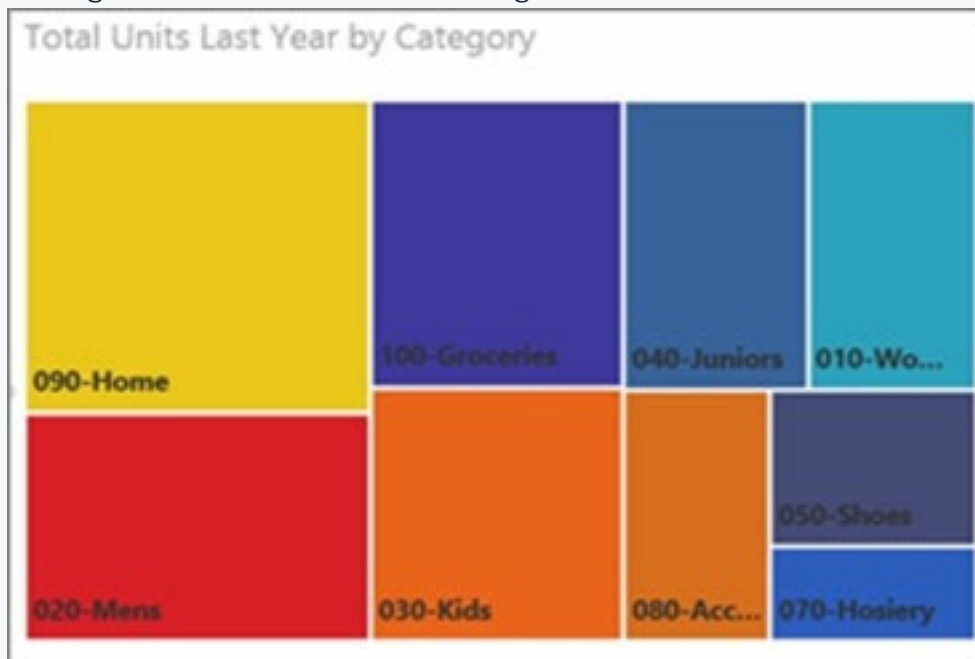
Scatter

A chart that shows the relationship between two numerical values.

Explanation:

Box 1: Tree map -

Treemaps are charts of colored rectangles, with size representing value. They can be hierarchical, with rectangles nested within the main rectangles.



Box 2: Key influencer -

A key influencer chart displays the major contributors to a selected result or value.

Box 3: Scatter -

Scatter and Bubble charts display relationships between 2 (scatter) or 3 (bubble) quantitative measures -- whether or not, in which order, etc.

Question: 16

You need to create an Azure Storage account.

Data in the account must replicate outside the Azure region automatically.
Which two types of replication can you use for the storage account? Each correct answer presents a complete solution.

NOTE: Each correct selection is worth one point.

- A. zone-redundant storage (ZRS)
- B. read-access geo-redundant storage (RA-GRS)
- C. locally-redundant storage (LRS)
- D. geo-redundant storage (GRS)

Answer: BD

Explanation:

D: Azure Storage offers two options for copying your data to a secondary region:

- ⇒ Geo-redundant storage (GRS)
- ⇒ Geo-zone-redundant storage (GZRS)

B: With GRS or GZRS, the data in the secondary region isn't available for read or write access unless there is a failover to the secondary region. For read access to the secondary region, configure your storage account to use read-access geo-redundant storage (RA-GRS) or read-access geo-zone-redundant storage (RA-GZRS).

Reference:

<https://docs.microsoft.com/en-us/azure/storage/common/storage-redundancy#redundancy-in-a-secondary-region>

Question: 17

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HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Platform as a service (PaaS) database offerings in Azure require less setup and configuration effort than infrastructure as a service (IaaS) database offerings.	☐	☐
Platform as a service (PaaS) database offerings in Azure provide end users with the ability to control and update the operating system version.	☐	☐
All relation and non-relational platform as a service (PaaS) database offerings in Azure can be paused to reduce costs.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Platform as a service (PaaS) database offerings in Azure require less setup and configuration effort than infrastructure as a service (IaaS) database offerings.	☒	☐
Platform as a service (PaaS) database offerings in Azure provide end users with the ability to control and update the operating system version.	☐	☒
All relation and non-relational platform as a service (PaaS) database offerings in Azure can be paused to reduce costs.	☐	☒

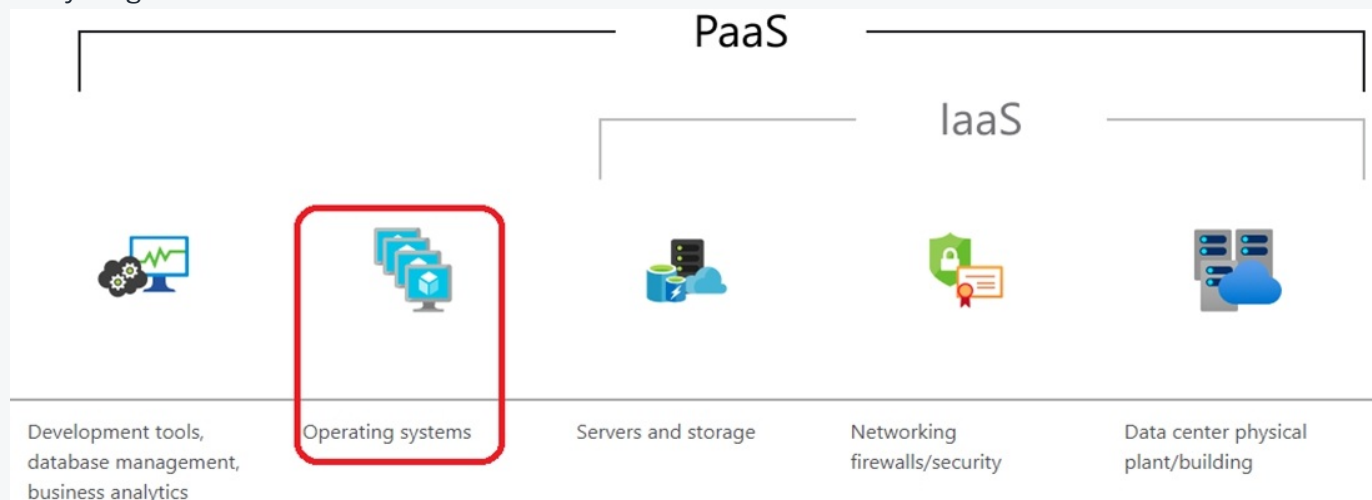
Explanation:

Box 1: Yes -

Like IaaS, PaaS includes infrastructure " servers, storage, and networking " but also middleware, development tools, business intelligence (BI) services, database management systems, and more. PaaS is designed to support the complete web application lifecycle: building, testing, deploying, managing, and updating. PaaS allows you to avoid the expense and complexity of buying and managing software licenses, the underlying application infrastructure and middleware, container orchestrators such as Kubernetes, or the development tools and other resources

Box 2: No -

You manage the applications and services you develop, and the cloud service provider typically manages everything else.



Box 3: No -

There really is no way to pause / stop billing for your Azure SQL Database.

Microsoft's official answer "Yes, you can export your database. Delete the Azure SQL database and that will pause billing. Then when you need it you can create a new database and import your previously exported DB."

Reference:

<https://azure.microsoft.com/en-us/overview/what-is-paas>

Question: 18

CertyIQ

Which statement is an example of Data Manipulation Language (DML)?

- A. REVOKE
- B. DISABLE

C. INSERT

D. GRANT

Answer: C

Explanation:

Data Manipulation Language (DML) affect the information stored in the database. Use these statements to insert, update, and change the rows in the database.

BULK INSERT -

-
- ⇒ DELETE
- ⇒ INSERT
- ⇒ SELECT
- ⇒ UPDATE
- ⇒ MERGE

Reference:

<https://docs.microsoft.com/en-us/sql/t-sql/statements/statements>

Question: 19

CertyIQ

You have a SQL query that combines customer data and order data. The query includes calculated columns. You need to create a database object that would allow other users to rerun the same SQL query. What should you create?

- A. an index
- B. a view
- C. a scalar function
- D. a table

Answer: B

Explanation:

A view is a virtual table whose contents are defined by a query. A view acts as a filter on the underlying tables referenced in the view. The query that defines the view can be from one or more tables or from other views in the current or other databases.

Reference:

<https://docs.microsoft.com/en-us/sql/relational-databases/views/views>

Question: 20

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HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A key/value data store is optimized for

	▼
enforcing constraints	
simple lookups	
table joins	
transactions	

Answer:

Answer Area

A key/value data store is optimized for

	▼
enforcing constraints	
simple lookups	
table joins	
transactions	

Explanation:

Box 1: simple lookups -

A key/value store associates each data value with a unique key. Most key/value stores only support simple query, insert, and delete operations. To modify a value (either partially or completely), an application must overwrite the existing data for the entire value. In most implementations, reading or writing a single value is an atomic operation.

An application can store arbitrary data as a set of values. Any schema information must be provided by the application. The key/value store simply retrieves or stores the value by key.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 21

CertyIQ

DRAG DROP -

Match the types of data to the appropriate Azure data services.

To answer, drag the appropriate data type from the column on the left to its service on the right. Each data type may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Data Types

Image files
Key/value pairs
Relationships between employees

Answer Area

Data type	Azure Blob storage
Data type	Azure Cosmos DB Gremlin API
Data type	Azure Table storage

Answer:

Data Types	Answer Area
Image files	Image files Azure Blob storage
Key/value pairs	Relationships between employees Azure Cosmos DB Gremlin API
Relationships between employees	Key/value pairs Azure Table storage

Explanation:

1. Image files = Azure Blob storage

2. Relationship between employees = Azure Cosmos DB Gremlin API

3. Key/value pairs = Azure Table Storage

Box 1: Image files -

Azure Blob storage is suitable for image files.

Box 2: Relationship between employees

Azure Cosmos DB Gremlin API

Box 3:Key/value pairs -

Azure Table Storage

Reference:

<https://docs.microsoft.com/en-us/azure/storage/tables/table-storage-design-modeling>

Question: 22

CertyIQ

DRAG DROP -

Match the Azure Data Lake Storage Gen2 terms to the appropriate levels in the hierarchy.

To answer, drag the appropriate term from the column on the left to its level on the right. Each term may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Terms

Azure Storage account

File share

Container

Answer Area

Azure Resource Group

Term

Term

Folders

Files

Answer:

Terms

Azure Storage account

File share

Container

Answer Area

Azure Resource Group

Azure Storage account

Container

Folders

Files

Explanation:

Box 1: Azure Storage account -

Azure file shares are deployed into storage accounts, which are top-level objects that represent a shared pool of storage.

Box 2: Container -

Reference:

<https://docs.microsoft.com/en-us/azure/storage/files/storage-how-to-create-file-share>

Question: 23

CertyIQ

What are two characteristics of real-time data processing? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

- A. Data is processed periodically
- B. Low latency is expected
- C. High latency is acceptable

D. Data is processed as it is created

Answer: BD

Explanation:

Real time processing deals with streams of data that are captured in real-time and processed with minimal latency to generate real-time (or near-real-time) reports or automated responses.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/big-data/real-time-processing>

Question: 24

CertyIQ

DRAG DROP -

Match the Azure Data Factory components to the appropriate descriptions.

To answer, drag the appropriate component from the column on the left to its description on the right. Each component may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Components	Answer Area
Dataset	Component A representation of data structures within data stores
Linked service	Component The information used to connect to external resources
Mapping data flow	Component A logical grouping of activities that performs a unit of work and can be scheduled
Pipeline	

Answer:

Components	Answer Area
Dataset	Dataset A representation of data structures within data stores
Linked service	Linked service The information used to connect to external resources
Mapping data flow	
Pipeline	Pipeline A logical grouping of activities that performs a unit of work and can be scheduled

Explanation:

Box 1: Dataset -

Datasets must be created from paths in Azure datastores or public web URLs, for the data to be accessible by Azure Machine Learning.

Box 2: Linked service -

Linked services are much like connection strings, which define the connection information needed for Data Factory to connect to external resources.

Box 3: Pipeline -

A pipeline is a logical grouping of activities that together perform a task.

Reference:

<https://k21academy.com/microsoft-azure/dp-100/datastores-and-datasets-in-azure/> <https://docs.microsoft.com/en-us/azure/data-factory/concepts-linked-services> <https://docs.microsoft.com/en-us/azure/data-factory/concepts-pipelines-activities>

Question: 25

CertyIQ

DRAG DROP -

Match the types of workloads to the appropriate scenarios.

To answer, drag the appropriate workload type from the column on the left to its scenario on the right. Each workload type may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Workload Types

Answer Area

Batch

Streaming

Workload type

Data for a product catalog will be loaded every 12 hours to a data warehouse.

Workload type

Thousands of data sets per second for online purchases will be loaded into a data warehouse in real time

Workload type

Updates to inventory data will be loaded to a data warehouse every 1 million transactions.

Answer:

Workload Types

Answer Area

Batch

Streaming

Batch

Data for a product catalog will be loaded every 12 hours to a data warehouse.

Streaming

Thousands of data sets per second for online purchases will be loaded into a data warehouse in real time

Batch

Updates to inventory data will be loaded to a data warehouse every 1 million transactions.

Explanation:

Box 1: Batch -

Batch processing refers to the processing of blocks of data that have already been stored over a period of time.

Box 2: Streaming -

Stream processing is a big data technology that allows us to process data in real-time as they arrive and detect conditions within a small period of time from the point of receiving the data. It allows us to feed data into analytics tools as soon as they get generated and get instant analytics results.

Box 3: Batch -

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/batch-processing>

Question: 26

DRAG DROP -

Your company plans to load data from a customer relationship management (CRM) system to a data warehouse by using an extract, load, and transform (ELT) process.

Where does data processing occur for each stage of the ELT process? To answer, drag the appropriate locations to the correct stages. Each location may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.

Select and Place:

Locations

Answer Area

A standalone data analysis tool

Extract:

The CRM system

Load:

The data warehouse

Transform:

Answer:

Locations

Answer Area

Extract:

The CRM system

Load:

The data warehouse

Transform:

The data warehouse

Explanation:

Box 1: The CRM system -

Data is extracted from the CRM system.

Box 2: The data warehouse -

Data is loaded to the data warehouse.

Box 3: The data warehouse -

"Extract, load, and transform (ELT) differs from ETL solely in where the transformation takes place. In the ETL pipeline, the transformation occurs in the target data store. Instead of using a separate transformation engine, the processing capabilities of the target data store are used to transform data (...) In practice, the TARGET data store is a DATA WAREHOUSE using either a Hadoop cluster (using Hive or Spark) or a SQL dedicated pools on Azure Synapse Analytics" (<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/etl>) = DW for the Transform.

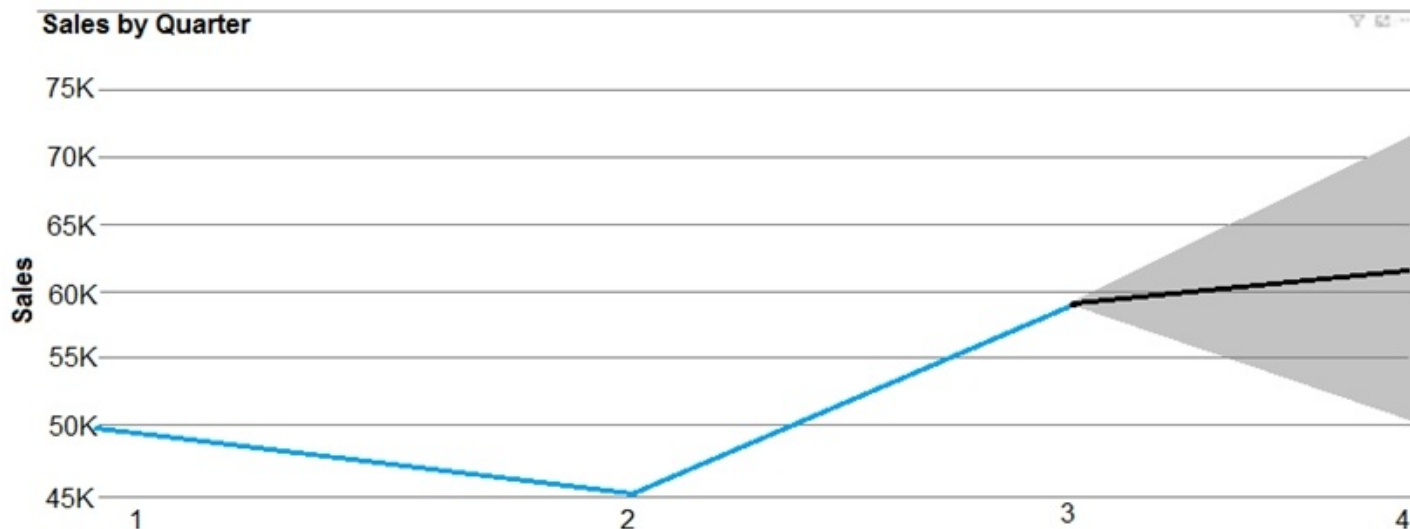
Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/etl>

Question: 27

CertyIQ

Your company recently reported sales from the third quarter. You have the chart shown in the following exhibit.



Which type of analysis is shown in the fourth quarter?

- A. predictive
- B. prescriptive
- C. descriptive
- D. diagnostic

Answer: A

Explanation:

Predictive, to answer the question: What will happen?

			
Descriptive	Diagnostic	Predictive	Prescriptive
<i>What's happening?</i>	<i>Why's happening?</i>	<i>What will happen?</i>	<i>What actions should we take?</i>

Reference:

<https://demand-planning.com/2020/01/20/the-differences-between-descriptive-diagnostic-predictive-cognitive-analytics/> <https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/>

Question: 28

Which statement is an example of Data Manipulation Language (DML)?

- A. REVOKE
- B. DISABLE
- C. CREATE
- D. UPDATE

Answer: D

Explanation:

Data Manipulation Language (DML) affect the information stored in the database. Use these statements to insert, update, and change the rows in the database.

BULK INSERT -

DELETE -

INSERT -

SELECT -

UPDATE -

MERGE -

There's 3 type of Relational data SQL:

DDL (Data Definition Language) CREATE, DROP, ALTER, RENAME, COMMENT AND TRUNCATE.

DML (Data Manipulación Language) SELECT, INSERT INTO, DELETE, UPDATE.

DCL (Data Control Language) REVOKE & GRANT.

TCL (Transaccional Control Language) COMMIT & ROLLBACK

Reference:

<https://docs.microsoft.com/en-us/sql/t-sql/statements/statements>

Question: 29

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Creating closed caption text for audio files is an example of

	▼
cognitive	
descriptive	
predictive	
prescriptive	

analytics.

Answer:

Answer Area

Creating closed caption text for audio files is an example of analytics.

cognitive
descriptive
predictive
prescriptive

Explanation:

Descriptive, to answer the question: What's happening?



Descriptive
What's happening?



Diagnostic
Why's happening?



Predictive
What will happen?



Prescriptive
What actions should we take?

Note: Azure Media Indexer enables you to make content of your media files searchable and to generate a full-text transcript for closed captioning and keywords.

You can process one media file or multiple media files in a batch.

Reference:

<https://demand-planning.com/2020/01/20/the-differences-between-descriptive-diagnostic-predictive-cognitive-analytics/> <https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/> <https://docs.microsoft.com/en-us/azure/media-services/previous/media-services-index-content>

Question: 30

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A visualization that illustrates a university's current student enrollment per department is an example of analytics.

▼
cognitive
descriptive
predictive
prescriptive

Answer:

Answer Area

A visualization that illustrates a university's current student enrollment per department is an example of analytics.

▼
cognitive
descriptive
predictive
prescriptive

Explanation:

Generally speaking, data analytics comes in four types:

1. Descriptive, to answer the question: What's happening?
2. Diagnostic, to answer the question: Why's happening?
3. Predictive, to answer the question: What will happen?
4. Prescriptive, to answer the question: What actions should we take?



Descriptive
What's happening?



Diagnostic
Why's happening?



Predictive
What will happen?



Prescriptive
What actions should we take?

Reference:

<https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/>

Question: 31

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

An extract, load, and transform (ELT) process
requires

- a separate transformation engine.
- a target data store powerful enough to transform data.
- data that is fully processed before being loaded to the target data store.
- a data pipeline that includes a transformation engine.

Answer:**Answer Area**

An extract, load, and transform (ELT) process
requires

- a separate transformation engine.
- a target data store powerful enough to transform data.
- data that is fully processed before being loaded to the target data store.
- a data pipeline that includes a transformation engine.

Explanation:

With ELT, the data store used to perform the transformation is the same data store where the data is ultimately consumed.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/etl>**Question: 32**

You need to create an Azure resource to store data in Azure Table storage.

Which command should you run?

- A. az storage share create
- B. az storage account create
- C. az cosmosdb create
- D. az storage container create

Answer: B**Explanation:**

Due to no az storage table create in the selection. The best answer is B create storage account (first)

Question: 33

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

The Azure Cosmos DB

Core (SQL)
Gremlin
MongoDB
Table

 API enables the use of SELECT statements to

retrieve documents from Azure Cosmos DB.

Answer:

Answer Area

The Azure Cosmos DB

Core (SQL)
Gremlin
MongoDB
Table

 API enables the use of SELECT statements to

retrieve documents from Azure Cosmos DB.

Explanation:

Azure Cosmos DB SQL API accounts provide support for querying items using the Structured Query Language (SQL) syntax.

Reference:

<https://docs.microsoft.com/en-us/azure/cosmos-db/choose-api>

Question: 34

CertyIQ

You need to modify a view in a relational database by adding a new column.
Which statement should you use?

- A. MERGE
- B. ALTER
- C. INSERT
- D. UPDATE

Answer: B

Explanation:

"Modify a View" = Alter View

There's 3 type of Relational data SQL:

DDL (Data Definition Language) CREATE, DROP, ALTER, RENAME, COMMENT AND TRUNCATE.

DML (Data Manipulación Language) SELECT, INSERT INTO, DELETE, UPDATE.

DCL (Data Control Language) REVOKE & GRANT.

TCL (Transactional Control Language) COMMIT & ROLLBACK

Question: 35

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Extract, transform, and load (ETL) can reduce the transfer of sensitive data to destination systems.	☐	☐
Extract, load, and transform (ELT) transforms data by using a compute resource independent of the source system and destination system.	☐	☐
Extract, load, and transform (ELT) minimizes the time it takes to copy large volumes of data to destination systems.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Extract, transform, and load (ETL) can reduce the transfer of sensitive data to destination systems.	☒	☐
Extract, load, and transform (ELT) transforms data by using a compute resource independent of the source system and destination system.	☐	☒
Extract, load, and transform (ELT) minimizes the time it takes to copy large volumes of data to destination systems.	☒	☐

Explanation:

Sentence one is **yes** - You can reduce transfer sensitive data with transformations on data source.

Sentence two is **No** - ELT use the compute resource from destination system.

And for last **Yes**. ELT minimizes time with transformations are going realized on target data destination systems.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/data-guide/relational-data/etl>

Question: 36

CertyIQ

HOTSPOT -

You plan to deploy a PostgreSQL database to Azure.

Which hosting model corresponds to the available deployment options? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

PostgreSQL on Azure VMs:

☒ Infrastructure as a service (IaaS)

☒ Platform as a service (PaaS)

☒ Software as a service (SaaS)

Azure Database for PostgreSQL:

☒ Infrastructure as a service (IaaS)

☒ Platform as a service (PaaS)

☒ Software as a service (SaaS)

Answer:

Answer Area

PostgreSQL on Azure VMs:

Infrastructure as a service (IaaS)
Platform as a service (PaaS)
Software as a service (SaaS)

Azure Database for PostgreSQL:

Infrastructure as a service (IaaS)
Platform as a service (PaaS)
Software as a service (SaaS)

Explanation:

Que said VM --> Hence IaaS

so, IaaS AND PaaS is correct

Reference:

<https://azure.microsoft.com/en-us/overview/what-is-saas/>

<https://azure.microsoft.com/en-us/overview/what-is-paas/>

Question: 37

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Column family databases
Document databases
Graph databases
Key-value stores

_____ natively support the analysis of relationships between entities.

Answer:

Answer Area

▼
Column family databases
Document databases
Graph databases
Key-value stores

natively support the analysis of relationships between entities.

Explanation:

Graph databases can efficiently perform queries across the network of nodes and edges and analyze the relationships between entities.

<https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 38

CertyIQ

Which Azure storage solution provides native support for POSIX-compliant access control lists (ACLs)?

- A. Azure Table storage
- B. Azure Data Lake Storage
- C. Azure Queue storage
- D. Azure Files

Answer: B

Explanation:

Reference:

<https://docs.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-access-control>

Question: 39

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Relational data is stored in

- a file system as unstructured data.
- a hierarchal folder structure.
- a tabular form of rows and columns.
- comma-separated value (CSV) files.

Answer:

Answer Area

Relational data is stored in

- a file system as unstructured data.
- a hierarchal folder structure.
- a tabular form of rows and columns.
- comma-separated value (CSV) files.

Question: 40

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

- A view
- A clustered index
- A stored procedure
- A nonclustered index

physically sorts the data in a table based on the values in a specified column.

Answer:

Answer Area

A view
A clustered index
A stored procedure
A nonclustered index

physically sorts the data in a table based on the values in a specified column.

Explanation:

Answer is B, Clustered indexes.

<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/clustered-and-nonclustered-indexes-described?view=sql-server-ver15&viewFallbackFrom=sql-server-ver15%20%20Previous%20QuestionsNext%20Questions>

Reference:

<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/clustered-and-nonclustered-indexes-described?view=sql-server-ver15>

Question: 41

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

A block of code that runs in a database is called

a stored procedure.
a table.
a view.
an index.

Answer:

Answer Area

A block of code that runs in a database is called

a stored procedure.
a table.
a view.
an index.

Explanation:

Reference:

<https://docs.microsoft.com/en-us/sql/relational-databases/stored-procedures/stored-procedures-database-engine?view=sql-server-ver15>

Question: 42

CertyIQ

DRAG DROP -

Match the types of analytics that can be used to answer the business questions.

To answer, drag the appropriate analytics type from the column on the left to its question on the right. Each analytics type may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Analytics Types

Answer Area

Cognitive

Why did sales increase last month?

Diagnostic

Provide specific guidance on how to allocate current budget to buy different inventory items.

Descriptive

Predictive

Detect images of the company's logo included in PDF documents

Prescriptive

Answer:

Analytics Types

Answer Area

- Cognitive
- Diagnostic
- Descriptive
- Predictive
- Prescriptive

Why did sales increase last month?

Provide specific guidance on how to allocate current budget to buy different inventory items.

Detect images of the company's logo included in PDF documents

Diagnostic

Prescriptive

Cognitive

ties.

Explanation:

Box 1: Diagnostic -

Diagnostic Analytics: At this stage you can begin to answer some of those why questions. Historical data can begin to be measured against other data to answer the question of why something happened in the past. This is the process of gathering and interpreting different data sets to identify anomalies, detect patterns, and determine relationships.

Box 2: Prescriptive -

Prescriptive analytics is a combination of data, mathematical models, and various business rules to infer actions to influence future desired outcomes.

Predictive analytics, broadly speaking, is a category of business intelligence that uses descriptive and predictive variables from the past to analyze and identify the likelihood of an unknown future outcome

Box 3: Cognitive -

Reference:

<https://demand-planning.com/2020/01/20/the-differences-between-descriptive-diagnostic-predictive-cognitive-analytics/> <https://azure.microsoft.com/en-us/blog/answering-whats-happening-whys-happening-and-what-will-happen-with-iot-analytics/>

Question: 43

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

	▼
A heap	
A stored procedure	
A view	
An index	

presents content defined by a query.

Answer:

Answer Area

	▼
A heap	
A stored procedure	
A view	
An index	

presents content defined by a query.

Question: 44

CertyIQ

Which type of database is Azure Database for PostgreSQL?

- A. Platform as a service (PaaS)
- B. Infrastructure as a service (IaaS)
- C. Microsoft SQL Server
- D. on-premises

Answer: A

Explanation:

Correct answer: A, PAAS

Reference:

<https://docs.microsoft.com/en-us/azure/postgresql/overview-postgres-choose-server-options>

Question: 45

Which storage solution supports access control lists (ACLs) at the file and folder level?

- A. Azure Data Lake Storage
- B. Azure Queue storage
- C. Azure Blob storage
- D. Azure Cosmos DB

Answer: A

Explanation:

Azure Data Lake Storage Gen2 implements an access control model that supports both Azure role-based access control (Azure RBAC) and POSIX-like access control lists (ACLs).

Reference:

<https://docs.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-access-control>

Question: 46

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

A

	▼
line chart	
matrix	
scatter	
treemap	

 is a chart of colored rectangles. The size of the rectangles represent

the relative value of each item. The chart can be hierarchical, displaying data as a set of rectangles nested within the main rectangle.

Answer:

Answer Area

A

	▼
line chart	
matrix	
scatter	
treemap	

 is a chart of colored rectangles. The size of the rectangles represent

the relative value of each item. The chart can be hierarchical, displaying data as a set of rectangles nested within the main rectangle.

Explanation:

A treemap chart divides the chart area into rectangles that represent the different levels and relative sizes of the data hierarchy.

Reference:

<https://docs.microsoft.com/en-us/sql/reporting-services/report-design/tree-map-and-sunburst-charts-in-reporting-services?view=sql-server-ver15>

Question: 47

CertyIQ

What is a characteristic of batch processing?

- A. The data ingested during batch processing must be processed as soon as the data is received.
- B. Large datasets must be split into batches of less than 1 GB before the data can be processed.
- C. There is a noticeable time delay between ingesting data and obtaining the data processing results.
- D. Batch processing can only process data that is structured.

Answer: C

Explanation:

C. Batch processing can process both structured and unstructured data.

Question: 48

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

In a SQL statement, ▼ are examples of aggregate functions.

FROM and WHERE
GROUP BY and ORDER BY
JOIN and MERGE
MAX and MIN

Answer:

Answer Area

In a SQL statement, ▼ are examples of aggregate functions.

FROM and WHERE
GROUP BY and ORDER BY
JOIN and MERGE
MAX and MIN

Explanation:

An aggregate function performs a calculation on a set of values, and returns a single value. The following are aggregate functions:

- ⇒ APPROX_COUNT_DISTINCT
- ⇒ AVG

- CHECKSUM_AGG
- COUNT
- COUNT_BIG

GROUPING -

-
- GROUPING_ID
- MAX
- MIN
- STDEV
- STDEVP
- STRING_AGG
- SUM
- VAR
- VARP

Reference:

<https://docs.microsoft.com/en-us/sql/t-sql/functions/aggregate-functions-transact-sql?view=sql-server-ver15>

Question: 49

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
A job that processes sales data once daily is an example of a batch workload.	☐	☐
A job that calculates a rolling average temperature reading is an example of a streaming workload.	☐	☐
A job that calculates average revenue per product for the last month is an example of a streaming workload.	☐	☐

Answer:

Answer Area

Statements	Yes	No
A job that processes sales data once daily is an example of a batch workload.	☒	☐
A job that calculates a rolling average temperature reading is an example of a streaming workload.	☒	☐
A job that calculates average revenue per product for the last month is an example of a streaming workload.	☐	☒

Explanation:

Box 1: Yes -

Batch processing refers to processing of high volume of data in batch within a specific time span. It processes large volume of data all at once. Batch processing is used when data size is known and finite. It takes little

longer time to processes data.

Box 2: Yes -

Stream processing refers to processing of continuous stream of data immediately as it is produced. It analyzes streaming data in real time. Stream processing is used when the data size is unknown and infinite and continuous.

Box 3: No -

Reference:

<https://www.geeksforgeeks.org/difference-between-batch-processing-and-stream-processing/>

Question: 50

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

In a relational database, each row in a table has

	▼
a different set of columns	
a key-value pair	
the same set of columns	
unstructured data	

Answer:

Answer Area

In a relational database, each row in a table has

	▼
a different set of columns	
a key-value pair	
the same set of columns	
unstructured data	

Explanation:

The same set of columns.

In relational databases, a row is a data record within a table. Each row, which represents a complete record of specific item data, holds different data within the same structure. A row is occasionally referred to as a tuple.

Incorrect:

Not: a key value pair.

Unlike relational databases, key-value databases do not have a specified structure. Relational databases store data in tables where each column has an assigned data type. Key-value databases are a collection of key-value pairs that are stored as individual records and do not have a predefined data structure.

Reference:

<https://www.techopedia.com/definition/4425/database-row>

<https://www.techtarget.com/searchdatamanagement/tip/NoSQL-database-types-explained-Key-value-store>

Question: 51

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Normalizing a database increases the throughput of writing transactions.	☐	☐
Analytics systems are more normalized than transactional systems.	☐	☐
Normalizing a database results in queries that require more joins.	☐	☐

Answer:**Answer Area**

Statements	Yes	No
Normalizing a database increases the throughput of writing transactions.	☐	☒
Analytics systems are more normalized than transactional systems.	☐	☒
Normalizing a database results in queries that require more joins.	☒	☐

Explanation:

Box 1: No -

Database normalization is the process of restructuring a relational database in accordance with a series of so-called normal forms in order to reduce data redundancy and improve data integrity.

Full normalisation will generally not improve performance, in fact it can often make it worse but it will keep your data duplicate free.

Box 2: No -

Analytics systems are denormalized to increase performance.

Transactional database systems are normalized to increase data consistency.

Box 3: Yes -

Transactional database systems are more normalized and requires more joins.

Reference:

Question: 52

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

	▼	natively support the analysis of relationships between entities
Column family databases		
Document databases		
Graph databases		
Key-value stores		

Answer:

Answer Area

	▼	natively support the analysis of relationships between entities
Column family databases		
Document databases		
Graph databases		
Key-value stores		

Explanation:

A graph database stores two types of information, nodes and edges. Edges specify relationships between nodes. Nodes and edges can have properties that provide information about that node or edge, similar to columns in a table. Edges can also have a direction indicating the nature of the relationship.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 53

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

A key/value data store is optimized for

	▼
enforcing constraints.	
simple lookups.	
table joins.	

Answer:

Answer Area

A key/value data store is optimized for

	▼
enforcing constraints.	
simple lookups.	
table joins.	

Explanation:

A key/value store associates each data value with a unique key.

Key/value stores are highly optimized for applications performing simple lookups, but are less suitable if you need to query data across different key/value stores.

Key/value stores are also not optimized for querying by value.

A single key/value store can be extremely scalable, as the data store can easily distribute data across multiple nodes on separate machines.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 54

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

In an infrastructure as a service (IaaS) instance of Microsoft SQL Server on Azure, you manage the

	▼
elastic pool	
MySQL server	
PostgreSQL server	
virtual machine	

that hosts SQL Server

Answer:

Answer Area

In an infrastructure as a service (IaaS) instance of Microsoft SQL Server on Azure, you manage the

	▼
elastic pool	
MySQL server	
PostgreSQL server	
virtual machine	

that hosts SQL Server

Explanation:

SQL Server on Azure VM -

SQL Server on Azure VM falls into the industry category Infrastructure-as-a-Service (IaaS) and allows you to run SQL Server inside a fully managed virtual machine (VM) in Azure.

The most significant difference from SQL Database and SQL Managed Instance is that SQL Server on Azure Virtual Machines allows full control over the database engine. You can choose when to start maintenance/patching, change the recovery model to simple or bulk-logged, pause or start the service when needed, and you can fully customize the SQL Server database engine. With this additional control comes the added responsibility to manage the virtual machine.

Reference:

<https://docs.microsoft.com/en-us/azure/azure-sql/azure-sql-iaas-vs-paas-what-is-overview?view=azuresql>

Question: 55

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

Varying fields for each entity in a JSON document is an example of

	▼
relational data.	
semi-structured data.	
structured data.	
unstructured data.	

Answer:

Answer Area

Varying fields for each entity in a JSON document is an example of

	▼
relational data.	
semi-structured data.	
structured data.	
unstructured data.	

Explanation:

For some semi-structured data formats (e.g. JSON), data sets are frequently a simple concatenation of multiple documents.

Reference:

<https://docs.snowflake.com/en/user-guide/semistructured-considerations.html>

Question: 56

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

A JSON document is an example of

	▼
graph data.	
relational data.	
semi-structured data.	
unstructured data.	

Answer:

Answer Area

A JSON document is an example of

	▼
graph data.	
relational data.	
semi-structured data.	
unstructured data.	

Explanation:

Semi-structured data -

Semi-structured data (e.g., JSON, CSV, XML) is the bridge between structured and unstructured data. It does not have a predefined data model and is more complex than structured data, yet easier to store than unstructured data.

Reference:

<https://www.ibm.com/cloud/blog/structured-vs-unstructured-data>

Question: 57

CertyIQ

Which property of a transactional workload guarantees that each transaction is treated as a single unit that either succeeds completely or fails completely?

A. atomicity

B. isolation

- C. durability
- D. consistency

Answer: A

Explanation:

The purpose of making transactions atomic is to prevent different transactions from interfering with one another. This can only happen if more than one user process is trying to access the database at the same time, as when a server allows several clients to use it concurrently. The simplest way to enforce atomicity is for the DBMS to refuse to start any transaction until the previous one has committed.

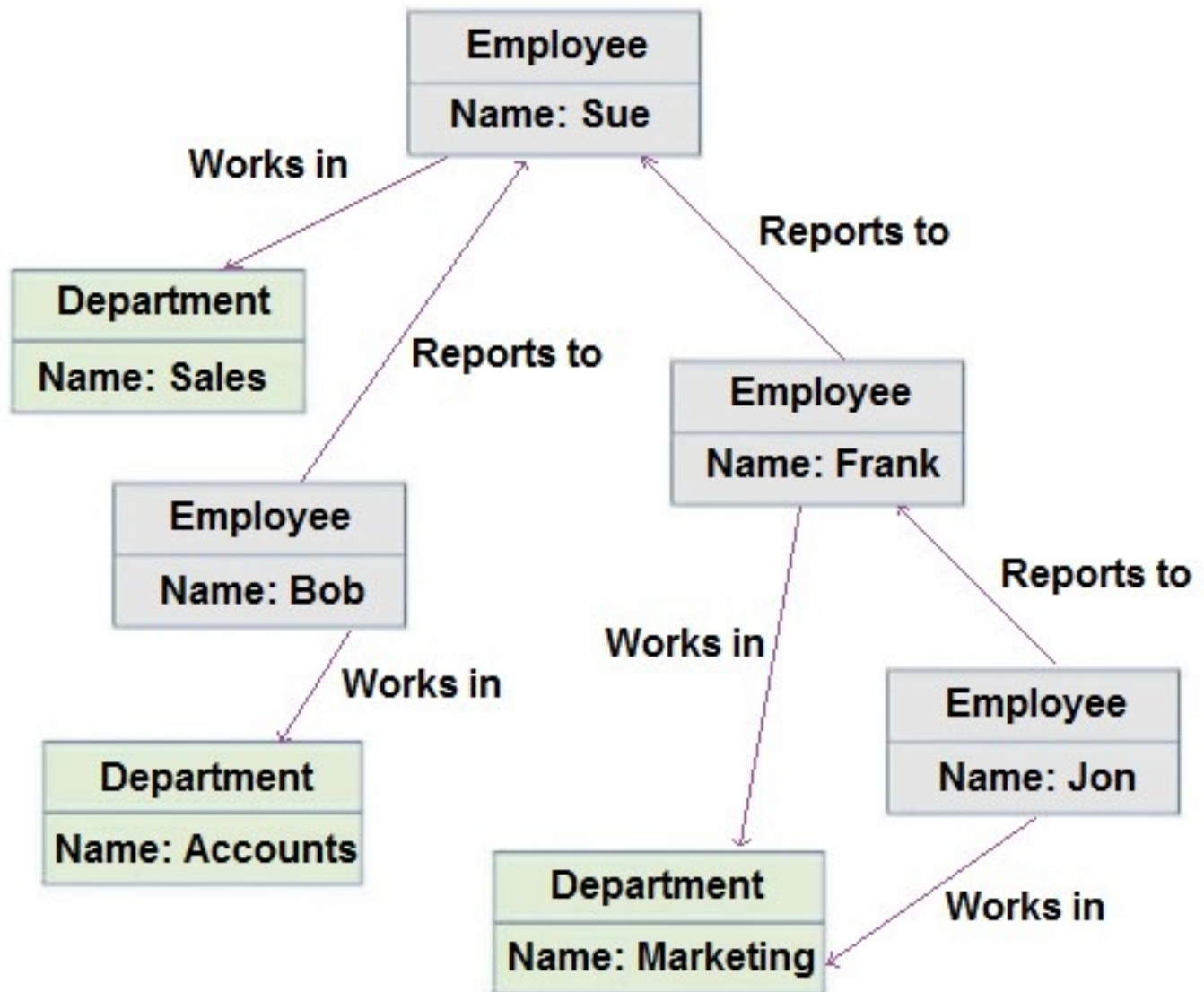
Reference:

<https://www.sciencedirect.com/topics/computer-science/atomic-transaction>

Question: 58

CertyIQ

You have a data store that has the structure shown in the following exhibit.



Which type of data store is this?

- A. key/value
- B. object data
- C. graph
- D. time series

Answer: C

Explanation:

A graph database stores two types of information, nodes and edges. Edges specify relationships between nodes. Nodes and edges can have properties that provide information about that node or edge, similar to columns in a table. Edges can also have a direction indicating the nature of the relationship.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 59

CertyIQ

Which type of database contains nodes and edges?

- A. graph
- B. key/value
- C. columnar
- D. time series

Answer: A

Explanation:

A graph database stores two types of information, nodes and edges. Edges specify relationships between nodes. Nodes and edges can have properties that provide information about that node or edge, similar to columns in a table. Edges can also have a direction indicating the nature of the relationship.

Reference:

<https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

Question: 60

CertyIQ

HOTSPOT -

Select the answer that correctly completes the sentence.

Hot Area:

Answer Area

	▼
A data analyst	
A data engineer	
A data scientist	

is responsible for identifying which business rules must be applied to the data of a company.

Answer:

Answer Area

	▼
A data analyst	
A data engineer	
A data scientist	

is responsible for identifying which business rules must be applied to the data of a company.

Explanation:

A data analyst's primary skill set revolves around data acquisition, handling, and processing.

Incorrect:

* A data engineer requires an intermediate level understanding of programming to build thorough algorithms along with a mastery of statistics and math.

* A data scientist needs to be a master of both worlds. Data, stats, and math along with in-depth programming knowledge for Machine Learning and Deep Learning.

Reference:

<https://www.edureka.co/blog/data-analyst-vs-data-engineer-vs-data-scientist/>

Question: 61

CertyIQ

HOTSPOT -

To complete the sentence, select the appropriate option in the answer area.

Hot Area:

Answer Area

Relational data uses

	▼
collections	
rows	
keys	
partitions	

 to enforce relationships between different tables.

Answer:

Answer Area

Relational data uses

	▼
collections	
rows	
keys	
partitions	

 to enforce relationships between different tables.

Explanation:

Reference:

Question: 62

You have an inventory management database that contains the following table.

ProductName	Quantity
Product1	100
Product2	129
Product3	176

Which statement should you use in a SQL query to change the inventory quantity of Product1 to 270?

- A. INSERT
- B. MERGE
- C. UPDATE
- D. CREATE

Answer: C

Explanation:

ALTER - add/delete/modify column

UPDATE - update existing row

INSERT - Insert new row

Reference:

<https://docs.microsoft.com/en-us/sql/t-sql/queries/update-transact-sql?view=sql-server-ver15>

Question: 63

Your company needs to implement a relational database in Azure. The solution must minimize ongoing maintenance.

Which Azure service should you use?

- A. Azure HDInsight
- B. Azure SQL Database
- C. Azure Cosmos DB
- D. SQL Server on Azure Virtual Machines

Answer: B

Explanation:

Reference:

<https://azure.microsoft.com/en-us/services/sql-database/#features>

Question: 64

You are writing a set of SQL queries that administrators will use to troubleshoot an Azure SQL database. You need to embed documents and query results into a SQL notebook. What should you use?

- A. Microsoft SQL Server Management Studio (SSMS)
- B. Azure Data Studio
- C. Azure CLI
- D. Azure PowerShell

Answer: B

Explanation:

B. Azure Data Studio is a cross-platform database tool that allows you to embed documents and query results into a SQL notebook. It supports Azure SQL databases, SQL Server, and other databases. It also has built-in support for SQL Notebooks, which allows you to mix markdown, code, and results in a single notebook. This makes it a great tool for creating documentation and troubleshooting guides for administrators to use when working with Azure SQL databases.

Reference:

<https://www.mssqltips.com/sqlservertip/5997/create-sql-server-notebooks-in-azure-data-studio/>

Question: 65

DRAG DROP -

Match the terms to the appropriate descriptions.

To answer, drag the appropriate term from the column on the left to its description on the right. Each term may be used once, more than once, or not at all.

NOTE: Each correct match is worth one point.

Select and Place:

Terms

Index
View
Table

Answer Area

	A database object that holds data
	A database object whose content is defined by a query
	A database object that helps improve the speed of data retrieval

Answer:

Terms

Index
View
Table

Answer Area

Table	A database object that holds data
View	A database object whose content is defined by a query
Index	A database object that helps improve the speed of data retrieval

Explanation:

Reference:

[https://en.wikipedia.org/wiki/Table_\(database\)](https://en.wikipedia.org/wiki/Table_(database))

[https://en.wikipedia.org/wiki/View_\(SQL\)](https://en.wikipedia.org/wiki/View_(SQL))
https://en.wikipedia.org/wiki/Database_index

Question: 66

CertyIQ

You have an e-commerce application that reads and writes data to an Azure SQL database. Which type of processing does the application use?

- A. stream processing
- B. batch processing
- C. Online Analytical Processing (OLAP)
- D. Online Transaction Processing (OLTP)

Answer: D

Explanation:

OLTP is designed to serve as a persistent data store for business or front-end applications. OLTP administers day to day transaction of an organization.

Online Transaction Processing (OLTP)

<https://learn.microsoft.com/en-us/training/modules/explore-core-data-concepts/5-transactional-data-processing?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wvl.azure-data-fundamentals-explore-core-data-concepts>

Reference:

<https://sqlwizard.blog/2020/03/15/sql-server-oltp-vs-olap/>

Question: 67

CertyIQ

When can you use an Azure Resource Manager template?

- A. to automate the creation of an interdependent group of Azure resources in a repeatable way
- B. to apply Azure policies for multi-tenant deployments
- C. to provision Azure subscriptions
- D. to control which services and feature administrators and developers can deploy from the Azure portal

Answer: A

Explanation:

You can automate deployments and use the practice of infrastructure as code. In code, you define the infrastructure that needs to be deployed

To implement infrastructure as code for your Azure solutions, use Azure Resource Manager templates (ARM templates). The template is a JavaScript Object

Notation (JSON) file that defines the infrastructure and configuration for your project. The template uses declarative syntax, which lets you state what you intend to deploy without having to write the sequence of programming commands to create it. In the template, you specify the resources to deploy and the properties for those resources.

Reference:

Question: 68

CertyIQ

You have an Azure SQL database that you access directly from the Internet.

You recently changed the public IP address of your computer.

After changing the IP address, you can no longer access the database. You can connect to other resources in Azure.

What is a possible cause of the issue?

- A. role-based access control (RBAC)
- B. Dynamic Host Configuration Protocol (DHCP)
- C. Domain Name Service (DNS)
- D. a database-level firewall

Answer: D

Explanation:

The Azure SQL Database firewall lets you decide which IP addresses may or may not have access to either your Azure SQL Server or your Azure SQL database.

When creating an Azure SQL Database, the firewall needs to be configured before anyone will be able to access the database. By default, no external access to your SQL Database will be allowed until you explicitly assign permission by creating a firewall rule.

Reference:

<https://www.sqlshack.com/configuring-the-azure-sql-database-firewall/>

Question: 69

CertyIQ

DRAG DROP -

Match the tools to the appropriate descriptions.

To answer, drag the appropriate tool from the column on the left to its description on the right. Each tool may be used once, more than once, or not at all.

Select and Place:

Tools	Answer Area
Azure Data Studio	Tool A graphical tool for managing SQL Server or Azure SQL databases that supports access, configuration, management, and administration tasks.
Microsoft SQL Server Data Tools (SSDT)	Tool A lightweight editor that can run on-demand SQL queries and view and save results as text, JSON, or Microsoft Excel files.
Microsoft SQL Server Management Studio (SSMS)	Tool A development tool for building Azure SQL databases, Microsoft SQL Server relational databases, SQL Server Analysis Services (SSAS) data models, SQL Server Integration Services (SSIS) packages, and SQL Server Reporting Services (SSRS) reports.

Answer:

Tools	Answer Area
	Microsoft SQL Server Management Studio (SSMS) A graphical tool for managing SQL Server or Azure SQL databases that supports access, configuration, management, and administration tasks.
	Azure Data Studio A lightweight editor that can run on-demand SQL queries and view and save results as text, JSON, or Microsoft Excel files.
	Microsoft SQL Server Data Tools (SSDT) A development tool for building Azure SQL databases, Microsoft SQL Server relational databases, SQL Server Analysis Services (SSAS) data models, SQL Server Integration Services (SSIS) packages, and SQL Server Reporting Services (SSRS) reports.

Explanation:

Box 1: Microsoft SQL Server Management Studio (SSMS)

SQL Server Management Studio (SSMS) is an integrated environment for managing any SQL infrastructure, from SQL Server to Azure SQL Database.

Box 2: Azure Data Studio -

Azure Data Studio offers a modern, keyboard-focused SQL coding experience that makes your everyday tasks easier with built-in features, such as multiple tab windows, a rich SQL editor, IntelliSense, keyword completion, code snippets, code navigation, and source control integration (Git). Run on-demand SQL queries, view and save results as text, JSON, or Excel. Edit data, organize your favorite database connections, and browse database objects in a familiar object browsing experience.

Box 3: Microsoft SQL Server Data Tools (SSDT)

SQL Server Data Tools (SSDT) is a modern development tool for building SQL Server relational databases, databases in Azure SQL, Analysis Services (AS) data models, Integration Services (IS) packages, and Reporting Services (RS) reports. With SSDT, you can design and deploy any SQL Server content type with the same ease as you would develop an application in Visual Studio.

Reference:

<https://docs.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms> <https://code.visualstudio.com/docs/supporting/FAQ> <https://docs.microsoft.com/en-us/sql/azure-data-studio/what-is-azure-data-studio> <https://docs.microsoft.com/en-us/sql/ssdt/download-sql-server-data-tools-ssdt>

Question: 70

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Relational database tables contain columns and rows	☐	☐
Indexes in a relational database describe the data types in a table	☐	☐
A database view is a virtual table whose content is defined by a query	☐	☐

Answer:

Answer Area

Statements	Yes	No
Relational database tables contain columns and rows	☒	☐
Indexes in a relational database describe the data types in a table	☐	☒
A database view is a virtual table whose content is defined by a query	☒	☐

Explanation:

Box 1: Yes -

Tables are database objects that contain all the data in a database. In tables, data is logically organized in a row-and-column format similar to a spreadsheet.

Each row represents a unique record, and each column represents a field in the record.

Box 2: No -

An index is an on-disk structure associated with a table or view that speeds retrieval of rows from the table or view.

Box 3: Yes -

A view is a virtual table whose contents are defined by a query. Like a table, a view consists of a set of named columns and rows of data.

Reference:

<https://docs.microsoft.com/en-us/sql/relational-databases/tables/tables> <https://docs.microsoft.com/en-us/sql/relational-databases/indexes/clustered-and-nonclustered-indexes-described> <https://docs.microsoft.com/en-us/sql/relational-databases/views/views?view=sql-server-ver15>

Question: 71

CertyIQ

Which command-line tool can you use to query Azure SQL databases?

- A. sqlcmd
- B. bcp
- C. azdata
- D. Azure CLI

Answer: A

Explanation:

The sqlcmd utility lets you enter Transact-SQL statements, system procedures, and script files at the command prompt.

Incorrect Answers:

B: The bulk copy program utility (bcp) bulk copies data between an instance of Microsoft SQL Server and a data file in a user-specified format.

D: The Azure CLI is the defacto tool for cross-platform and command-line tools for building and managing Azure resources.

Reference:

Question: 72

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Azure SQL Database includes a managed backup service.	☐	☐
Azure SQL Database has built-in high availability.	☐	☐
Azure SQL Database can use Azure Defender.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Azure SQL Database includes a managed backup service.	☒	☐
Azure SQL Database has built-in high availability.	☒	☐
Azure SQL Database can use Azure Defender.	☒	☐

Explanation:

Box 1: Yes -

Box 2: Yes -

Box 3: Yes -

Azure Defender provides security alerts and advanced threat protection for virtual machines, SQL databases, containers, web applications, your network, and more.

Azure Defender provides security alerts and advanced threat protection for virtual machines, SQL databases, containers, web applications, your network, and more.

Reference:

<https://docs.microsoft.com/en-us/azure/azure-sql/database/sql-database-paas-overview> <https://azure.micros>

Question: 73

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
You can use Azure Data Studio to query a Microsoft SQL Server big data cluster.	☐	☐
You can use Microsoft SQL Server Management Studio (SSMS) to query an Azure Synapse Analytics data warehouse.	☐	☐
You can use MySQL Workbench to query Azure Database for MariaDB databases.	☐	☐

Answer:

Answer Area

Statements	Yes	No
You can use Azure Data Studio to query a Microsoft SQL Server big data cluster.	☒	☐
You can use Microsoft SQL Server Management Studio (SSMS) to query an Azure Synapse Analytics data warehouse.	☒	☐
You can use MySQL Workbench to query Azure Database for MariaDB databases.	☒	☐

Explanation:

A) **Yes** from <https://docs.microsoft.com/en-us/sql/azure-data-studio/what-is?view=sql-server-ver15>

Use Azure Data Studio if you: Are connecting to a SQL Server 2019 big data cluster

B) **Yes** from <https://docs.microsoft.com/en-us/learn/modules/query-azure-sql-data-warehouse/4-query-dw-using-ssms>

C) **Yes** from <https://docs.microsoft.com/en-us/azure/mariadb/connect-workbench>

Reference:

<https://docs.microsoft.com/en-us/sql/big-data-cluster/connect-to-big-data-cluster?view=sql-server-ver15>
<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-query-ssms> <https://docs.microsoft.com/en-us/azure/mariadb/connect-workbench>

Question: 74

CertyIQ

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
Platform as a service (PaaS) database offerings in Azure provide built-in high availability.	☐	☐
Platform as a service (PaaS) database offerings in Azure provide configurable scaling options.	☐	☐
Platform as a service (PaaS) database offerings in Azure reduce the administrative overhead for managing hardware.	☐	☐

Answer:

Answer Area

Statements	Yes	No
Platform as a service (PaaS) database offerings in Azure provide built-in high availability.	☒	☐
Platform as a service (PaaS) database offerings in Azure provide configurable scaling options.	☒	☐
Platform as a service (PaaS) database offerings in Azure reduce the administrative overhead for managing hardware.	☒	☐

Explanation:

Reference:

<https://docs.microsoft.com/en-us/azure/azure-sql/database/sql-database-paas-overview>

Question: 75

CertyIQ

HOTSPOT -

You have the following SQL query.

```
INSERT INTO dbo.Products (ProductID, ProductName, Price, ProductDescription)
VALUES (1, 'Clamp', 12.48, 'Workbench clamp') ;
```

What are dbo.Products and ProductName? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

dbo.Products :

	▼
A column	
A database	
A table	
An index	

ProductName :

	▼
A column	
A database	
A table	
An index	

Answer:

Answer Area

dbo.Products :

	▼
A column	
A database	
A table	
An index	

ProductName :

	▼
A column	
A database	
A table	
An index	

Question: 76

HOTSPOT -

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Statements	Yes	No
You must apply operating system updates to Azure SQL databases regularly.	☐	☐
You need a Microsoft 365 subscription to create an Azure SQL database.	☐	☐
You can use existing Microsoft SQL Server licenses to reduce the cost of Azure SQL databases.	☐	☐

Answer:

Answer Area

Statements	Yes	No
You must apply operating system updates to Azure SQL databases regularly.	☐	☒
You need a Microsoft 365 subscription to create an Azure SQL database.	☐	☒
You can use existing Microsoft SQL Server licenses to reduce the cost of Azure SQL databases.	☒	☐

Explanation:

N/N/Y

Can I bring my on-premises or IaaS SQL Server license to Hyperscale? Yes, Azure Hybrid Benefit ...
(<https://docs.microsoft.com/en-us/azure/azure-sql/database/service-tier-hyperscale-frequently-asked-questions-faq>

<https://azure.microsoft.com/en-us/pricing/hybrid-benefit/#calculator>)

Azure Hybrid Benefit allows you to use SQL Server licenses with Software Assurance or qualifying

subscription licenses to pay a reduced base rate* for these products and services for SQL Server on Azure:

vCPU-based service tiers of Azure SQL Database (excluding serverless).

Azure SQL Managed Instance.

SQL Server in Azure Virtual Machines.

SQL Server Integration Services.

Reference:

<https://azure.microsoft.com/en-gb/blog/hot-patching-sql-server-engine-in-azure-sql-database/>

<https://azure.microsoft.com/en-us/services/sql-database/#product-overview>

Question: 77

CertyIQ

Which statement is an example of Data Definition Language (DDL)?

- A. SELECT
- B. JOIN
- C. UPDATE
- D. CREATE

Answer: D

Explanation:

Data Definition Language (DDL) statements defines data structures. Use these statements to create, alter, or drop data structures in a database. These statements include:

- ALTER
- Collations
- CREATE
- DROP
- DISABLE TRIGGER
- ENABLE TRIGGER
- RENAME
- UPDATE STATISTICS
- TRUNCATE TABLE
- INSERT

Reference:

<https://docs.microsoft.com/en-us/sql/t-sql/statements/statements>

Thank you

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